

CHAPTER 1: INTRODUCTION

1.1 Client / Organization Profile

Zeal Institute of Business Administration, Computer Application & Research (ZIBACAR) is a premier institute under Zeal Education Society, Pune. The institute is affiliated with Savitribai Phule Pune University (SPPU), approved by AICTE, New Delhi, and recognized by DTE, Government of Maharashtra. It offers postgraduate programs such as Master of Computer Applications (MCA) with a strong emphasis on academic excellence and practical exposure.

The institution promotes project-based learning as an integral part of its curriculum. Each semester, students are required to complete academic projects under the supervision of faculty mentors. The project lifecycle includes multiple stages such as title finalization, synopsis submission, documentation preparation, presentations, review meetings, and final submission.

Currently, much of the project coordination process is handled manually using physical documents, spreadsheets, and informal communication channels. To improve efficiency, transparency, and systematic monitoring, the institute requires a centralized digital system to manage the complete academic project lifecycle.

The proposed Project Management System aims to fulfill this requirement by providing a structured, role-based, and web-based solution.

1.2 Need for the System

In the current academic workflow, project monitoring involves manual tracking of submissions, approvals, and deadlines. Faculty members must manage multiple students and maintain separate records for each stage of the project. Students often depend on email, messaging applications, or physical submissions to share documents.

This manual process creates several challenges:

- Lack of centralized monitoring system.
- Difficulty in tracking stage-wise progress.

- Increased administrative workload for the Project Coordinator.
- Communication gaps regarding deadlines and approvals.
- Risk of misplaced or duplicate submissions.
- No automated notification mechanism.

As the number of students increases, managing academic projects becomes complex and time-consuming. Therefore, there is a strong need for a web-based system that automates batch creation, mentor assignment, stage management, document submission, approval workflow, and deadline monitoring.

The proposed Project Management System addresses these challenges by offering a structured and transparent digital platform for academic project tracking.

1.3 Scope & Feasibility of Work

Scope of the System

The Project Management System covers the following functionalities:

Admin (Project Coordinator)

- Create and manage batches.
- Assign faculty mentors to batches.
- Bulk upload students using Excel.
- Create, modify, and reorder project stages.
- Set expected submission deadlines.
- Monitor overall project progress.

Faculty (Mentor)

- View assigned students only.
- Review submitted documents.
- Approve or reject submissions.
- Provide remarks and suggestions.
- Track student progress stage-wise.

Student (Mentee)

- View assigned mentor.
- View stage timeline and deadlines.
- Upload documents (PDF/DOC/PPT).
- View status (Approved / Rejected / Late).
- Receive notifications.

General System Scope

- Role-based access control.
- Secure authentication system.
- Deadline tracking with late submission detection.
- Email notification system.
- Responsive web interface.
- Cloud deployment capability.

Feasibility of Work

Operational Feasibility

The system is user-friendly and requires minimal technical knowledge. Students and faculty can operate it using a web browser without special training.

Technical Feasibility

The system is developed using open-source technologies such as:

- HTML, CSS, JavaScript
- Python (Flask)
- MongoDB

These technologies are widely supported and suitable for academic-level deployment.

Economic Feasibility

The project uses free and open-source tools. MongoDB (Community Edition) and Flask have no licensing cost. Deployment can be done using free hosting services. Therefore, the system is economically feasible.

Time Feasibility

Due to its modular architecture and defined scope, the system can be developed within the academic project timeline.

1.4 Operating Environment – Hardware & Software

Hardware Requirements

- Personal Computer or Laptop
- Minimum 4 GB RAM
- Minimum 20 GB Free Storage
- Stable Internet Connection

Software Requirements

- Operating System: Windows
- Backend Framework: Python (Flask)
- Database: MongoDB
- Frontend: HTML5, CSS3, JavaScript
- Development Tool: Visual Studio Code
- Database Tool: MongoDB Compass
- Web Browser: Chrome / Firefox / Edge

1.5 Architecture of the System

The Project Management System follows a Three-Tier Architecture:

1. Presentation Layer

- User Interface (HTML, CSS, JavaScript)
- Role-based dashboards (Admin, Faculty, Student)

2. Application Layer

- Flask backend
- Business logic
- Authentication & authorization
- Stage workflow management
- File handling

- Email notification system

3. Data Layer

- MongoDB database
- Collections:
 - Users
 - Batches
 - Stages
 - Submissions
 - Notifications

The system follows a modular and scalable architecture, allowing easy enhancement in the future.

1.6 Detailed Description of Technology Used

1. HTML5

Used to design the structural layout of webpages such as login, dashboard, submission forms, and management panels.

2. CSS3

Used for styling the user interface, implementing responsive design, and maintaining consistent layout across devices.

3. JavaScript (ES6)

Provides interactivity including:

- Form validation

- Sidebar toggle functionality
- Dynamic content rendering
- Status updates

4. Python (Flask Framework)

Flask serves as the backend framework. It handles:

- URL routing
- Authentication & session management
- Role-based access control
- File upload handling
- Communication with MongoDB
- Email integration

5. MongoDB

MongoDB is a NoSQL database used to store:

- User information
- Batch details
- Stage configuration
- Submission records
- Notification logs